



Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology

Course Code & Title : IT3501 Full Stack Web Development

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 4 / Implementing Express in Node.js
- **Course Outcome** : CO4
- **Topic Learning Outcome:** TLO15
- **Activity Chosen** : Think Pair Share
- **Justification:**
 - Think Pair Share activity is chosen to make the students to think about and implement Express in a Node.js application. The understanding of steps involved in setting up an Express server in Node.js is essential for building real-time applications
 - Student think individually about the prerequisites for setting up a basic Express app like installing dependencies, initializing the app etc. This activity makes the student to familiarize with handling different HTTP methods and creating dynamic routes in Express.js
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - Instructor explained about the Setting up of Express framework, handling HTTP methods and creating dynamic routes. After explaining the Express.js concepts, the think pair share activity was carried out in the class room for a total of 20 minutes.
 - Students were given 5 minutes to independently think of how to create routes in an Express.js application to handle various HTTP methods like GET, POST, PUT and DELETE as shown in Figure 1.
 - Then the students were asked to discuss the answers with their neighbours as shown in Figure 2.
 - Finally, the Students shared their answers based on Express.js based routing questions as shown in Figure 3.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO9	PO10	PSO3
CO4	3	2	2	3	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO5	PO9	PO10	PSO3
Justification for correlation	Students will be able to understand the concept of Express set up, handling routes in Express.js application	Students will be able to apply Express based routing with various HTTP methods in real time applications	Students will be able to write Simple web server programs using Express.js	Students will be able to use modern development environment like Express framework to develop modern web applications	As the student involved in the activity as team work skill improved	Students communication skill will be improved as they are facing audience	The Students will be to apply the Express based routing concept in developing web applications

- **Images / Screenshot of the practice:**

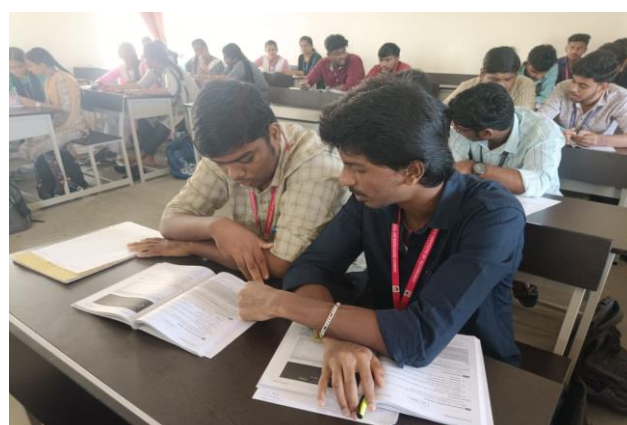


Fig 1: Students thinking about the given program Fig 2: Pair with neighbour and discuss answer



Fig 3: S. Aarthi & KV. Ashmitha share the answer to entire class

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students felt that this activity helps them to understand the Express.js based routing with different HTTP methods.
 - Students said that it will be very helpful to correct their mistakes while sharing the program solution to the other students.

- ❖ ***Benefit of the practice:***

- Students showed confidence in understanding basic routing syntax in Express.js
 - A few quieter students were more engaged during the "Pair" phase than in traditional lecture settings.

- ❖ ***Challenges faced in implementation:***

- Some students faced difficulty in implementing dynamic routes (e.g., `app.put('/users/:id')`) and understanding how to extract parameters using `req.params`
 - Debugging errors in route handling was challenging for some pairs.

I explained the `app.put('/users/:id')` dynamic route with an example and make them understand the topic.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2021-2022/3.VA_OCS752-ThinkPairShare.pdf
- <https://www.ritrjpm.ac.in/images/computer-science/2022-2023/TPS-RV-I-IT-PSPP.pdf>

Signature of Faculty Member

HOD